CP3402 Project

1. Purpose and Professional Context

> "What I look for in candidates … is curiosity and persistence. Everything else can be taught." — Keagan Chisnall, Engineering Manager, SafetyCulture

This group project is designed to develop **curiosity, persistence,** and **professional capability** through a realistic content management systems (CMS) project. You will work as a small development team to design, build, deploy, and document a professional WordPress website for a real client.

The emphasis of this assignment is **not only the final website** (product), but the **quality of the development process** used to produce it (process). You will be assessed on how you:

- work collaboratively,
- design and justify solutions,
- manage code and environments,
- document your work for handover, and
- reflect on your employability development.

This project mirrors real-world developer work: you will start without knowing everything, encounter obstacles, research alternatives, test approaches, and make justified decisions as a team.

2. Assessment Overview (50% of subject)

Component	Type	Weight
Project Website	Group	32
LinkedIn Project Report	Individual	8
Presentation	Group	8
Presentation Delivery Style	Individual	2
Total		**50**

3. Project Website (32%)

3.1 Task Overview

You will work in a **team of four** to design, develop, deploy, and document **one WordPress website** for a client.

The site must be built using a **custom WordPress theme**, developed from a recognised **starter theme**, and deployed using a **professional multi-environment workflow**.

You are expected to demonstrate:

- structured development practices,
- effective version control,
- local, staging, and production environments,
- clear, professional documentation, and
- meaningful and visible contribution from all team members.

3.2 Core Requirements (Checklist)

Your project **must include**:

- A custom WordPress theme developed from a recognised starter theme
- A functioning **local development environment**
- A **public staging site**
- A **public production site**
- A public **GitHub Classroom repository**
- A visible **project management board** using Trello or GitHub Projects
- Markdown documentation stored in the root of your repository
- A recorded presentation focused on your development and deployment **workflow**

A custom domain name is not required. Separate public servers are expected for staging and production

3.3 Limitations (Not Permitted)

The following are **not allowed**:

- Child themes
- WordPress.com
- Page builders or plugins that replace core WordPress functionality, e.g., Elementor, Nicepage
- Starting from a fully designed theme instead of a starter theme

Do not:

- Include WordPress core files in the GitHub repository (these can be ignored using .gitignore)
- Include secrets or passwords in your repository or project management board
- Hard-code site-specific content into your theme

3.4 Design Expectations

Your team is responsible for all design decisions, except where constrained by the client. Consider the usual aspects of good web design, including information architecture, interface design, and information design. All design decisions should be clearly connected to the client's goals and audience, as they would need to be justified to stakeholders in professional practice.

3.5 Theme Development Expectations

Your theme should:

- Follow WordPress theme development standards including coding styles
- Separate presentation from content
- Avoid fixed URLs, hard-coded IDs, or static content in templates
- Be structured so another developer could continue development
- Be reusable in any WordPress site by uploading and installing from zip file

Testing your theme using **WordPress Theme Test Data** is recommended to confirm reusability and correctness.

3.6 Content

- Content will be provided by the client
- You may refine wording, formatting, and structure
- You may add small amounts of content where appropriate
- Substantial rewriting is not expected

All additional media must comply with commercial-use licensing. The final site should be publication-ready.

3.7 Getting Started

Login to GitHub, then create or join your group's GitHub repo using this Classroom link: https://classroom.github.com/a/bHO-Eg7s

If you are the first in your group to create the GitHub repo, then use a repo name like “jcua-team34” where the number is the same as your team number - use jcua, jcub or jcus for your campus. This will give you, your team and your markers, access to your repository without having to add collaborators.

Create a channel in your IT@JCU Slack workspace, or a Discord server, and invite your lecturer to it.

Create a public project management board using Trello or GitHub Projects.

If you prefer to use different tools for communication or project management, these must be approved by your lecturer.

4. Workflow and Environments

A central learning objective of this project is to demonstrate a **professional development and deployment workflow** following core project management and DevOps practices.

Your project must include:

- Local development
- A staging environment distinct from production
- A live production environment

You are expected to:

- Research **multiple possible workflow approaches**
- Test alternatives where feasible
- Make an informed decision about your final workflow
- Clearly explain and justify that decision in your presentation

Higher levels of achievement are associated with workflows that are:

- clear and repeatable,
- well-documented,
- efficient and low-risk,
- appropriately automated where feasible.

5. Documentation (Markdown)

Documentation is assessed as a **professional handover artefact**, not an academic exercise. Write it so that someone else could realistically take over your project. All documentation must be written in Markdown and stored in the repository root. Write and update this documentation as you develop and deploy your site so that you can "eat your own dogfood".

Required Files

README.md

- Brief project overview
- Purpose of the repository
- Links to other documentation files

deployment.md

Explain your development and deployment workflow as a clear set of instructions so that a new team member could follow it:

- Local environment setup
- How changes are made and committed
- How changes are tested
- Deployment to staging
- Deployment to production
- Integration of project management and communication tools

Do not reproduce official WordPress or Docker documentation.

theme.md

Explain your theme to another developer so that they could efficiently take over development:

- Theme features
- Key files and structure
- Conventions used
- Important design decisions
- Anything non-obvious

site.md

Guide someone familiar with WordPress who is maintaining the site:

- How to add and edit content
- Pages vs posts
- Categories or content structure
- Plugin-specific processes
- Anything specific to this site's organisation

6. Project Management and Tools

Your team must:

- Use GitHub Classroom for repository management

- Maintain a **public** project management board using Trello or GitHub Projects
- Use appropriate communication tools using Slack or Discord.

Markers must be able to view:

- Repository activity and commit history
- Project management board
- Evidence of communication and collaboration

Every team member must be involved by communicating, using the board and making meaningful commits.

7. Hosting

Approved hosting providers listed in subject resources must be used.

- Free or student tiers are sufficient
- Staging and production should be on **separate public servers**

8. Peer Assessment

Peer assessment is **compulsory**. Failure to complete peer assessment will result in **zero marks** for the groupwork component of this assignment. Details for peer assessment will be provided separately. Each student will evaluate their own contribution and the contribution of other members.

Unequal contribution or ineffective teamwork may result in mark adjustments. Concerns about teamwork should be raised early with the lecturer.

9. Academic Integrity and Use of AI

You may:

- discuss ideas with other students,
- use external technical resources,
- use generative AI tools appropriately.

You remain responsible for:

- understanding all submitted code and documentation,
- ensuring compliance with WordPress standards and other expectations,
- avoiding copyright or licensing violations.

10. LinkedIn Project Report (8%, Individual)

Create a **LinkedIn Project** entry reflecting on this assignment. The intention of this LinkedIn report is to help *you* reflect on how your employability has improved during the subject – and to plan for this improvement throughout the subject. Completing this report thoughtfully will increase your visibility to potential employers and enrich what you learn in the subject.

Your report should:

- summarise the project,
- identify technologies and tools used,

- clearly describe your role and responsibilities,
- give specific examples of contributions,
- highlight skills developed,
- include proper links to the LinkedIn profiles of rest of your team (this is a feature of Projects on LinkedIn, not something you just put in the text
- fit in a LinkedIn Project (the size is constrained).

Write for a **professional audience**. Avoid vague skill lists. Tell concrete stories that demonstrate capability and growth.

Rather than writing this report quickly at the end of the project, work on it as part of your weekly learning journal throughout the project.

11. Presentation (8% Group + 2% Individual)

Your team must record a **7–10 minute** presentation focused on your **development and deployment workflow**.

The presentation must:

- demonstrate local, staging, and production environments,
- show a realistic update moving through the workflow,
- compare multiple workflow approaches considered,
- justify your workflow choices.

The presentation is **not** about the client or visual design. It is about your development and deployment process.

All group members must participate meaningfully. The individual delivery mark assesses clarity, professionalism, and engagement.

12. Submission

Group submission (one team member):

- Completed **project.html** template
- Zip file of your **theme**
- WordPress **export** of your site
- Presentation video

Individual submission (each team member):

- LinkedIn profile URL
- Text of your LinkedIn report as a PDF or Word file (remembering size constraint)

13. Common Pitfalls – Red Flags

The following issues occur most years and are a frequent reason for lost marks. Use this list as a **self-check throughout the project**, not just at the end.

Workflow & Environments

- **Only one environment in practice**: Claiming to have local, staging, and production, but only really using one. Markers look for evidence that changes genuinely flow through environments.
- **No realistic deployment demo**: Describing a workflow without *showing* a real update moving through it loses marks.

Version Control

- **Last-minute commits**: Large commits near the deadline suggest weak workflow and obscure individual contribution.
- **One person doing all the commits**: Immediately visible and may result in mark adjustments.
- **Poor commit messages**: Messages like "update" or "fix stuff" weaken evidence of professional practice.
- **Tracking the wrong files**: Including WordPress core files or uploads in the repository is a common and avoidable mistake.

Theme Development

- **Hard-coded content**: Embedding client text, IDs, or URLs directly into templates breaks reusability and is penalised.
- **Theme tied to one site**: If your theme cannot reasonably be reused on another site, it does not meet requirements.
- **Starting from the wrong base**: Using a near-complete theme instead of a starter theme, or using a child theme, fails core requirements.

Documentation

- **Documentation written at the end**: Leads to vague, incomplete, or inaccurate handover material.
- **Audience mismatch**: Writing for markers instead of future developers or maintainers reduces quality.
- **General rather than specific:** Documentation that describes workflow but is not able to be followed like a set of instructions misses the point.

Design & Content

- **Design without justification**: Visual choices not linked to goals and audience.
- **Treating content as filler**: Poor formatting or hierarchy undermines information design.

Presentation

- **Talking about the site instead of the workflow**: The presentation is assessed on *process*, not features or aesthetics.
- **No comparison of approaches**: Showing only one workflow without alternatives limits marks.
- **Unbalanced participation**: Dominance by one or two people affects group marks.

Teamwork & Process

- **Ignoring team issues**: Problems not raised early cannot be fixed at the end.
- **Unequal workload masked by goodwill**: Peer assessment will surface this.
- **Limited communication**: Minimal comments focused on "getting the assignment done" do not demonstrate professionalism.
- **Useless project management**: Project management board shows unhelpful task breakdown and allocation.

LinkedIn Project Report

- **Generic skill lists**: Vague claims without concrete examples score poorly.
- **Written for the lecturer, not employers**: The audience is a potential employer.
- **Leaving it until the last week**: The strongest reports are built gradually through reflection.

14. What Strong Teams Consistently Do Well – Green Flags

The following behaviours and practices are commonly seen in **high-performing teams** and are strong indicators of higher-quality learning and assessment outcomes. Use this section as a **positive checklist** alongside the Common Pitfalls (Red Flags).

Workflow & Environments

- **Environments are used intentionally**: Local is used for development, staging for testing and review, and production only for approved changes.
- **Changes move visibly through the workflow**: Commits, deployments, and updates clearly reflect a flow from local → staging → production.
- **Workflow decisions are justified**: Teams can clearly explain *why* they chose a particular approach and what alternatives they considered.
- **Automation is appropriate, reducing manual tasks**: The workflow is simple while still being professional and reliable.

Version Control

- **Regular, incremental commits**: Work is committed steadily throughout the project rather than in large bursts.
- **Clear commit messages**: Messages, in the imperative mood, describe changes, making the history easy to follow.
- **Shared ownership of the codebase**: All team members contribute meaningfully to theme code and documentation.

Theme Development

- **Separation of content and presentation**: Templates are flexible and reusable, with no site-specific assumptions baked in.

- **Testing for reusability**: Teams validate their theme using Theme Test Data or similar techniques.

Documentation

- **Documentation evolves with the project**: Files are updated as decisions are made, not written retrospectively.
- **Audience-aware writing**: Each document clearly targets its intended reader (developer vs site maintainer).
- **Explains decisions, not just steps**: Documentation captures rationale as well as process.
- **Documentation written as instructions**: Developers can follow the instructions without guesswork – clearly follows the dogfooding principle.

Design & Content

- **Design choices are goal-driven**: Visual and structural decisions are clearly linked to client goals and audience needs.
- **Content is treated as part of design**: Text, headings, and media are formatted deliberately for clarity and scanning.

Presentation

- **Demonstration-led presentation**: The workflow is shown in action, not just described.
- **Balanced participation**: All group members contribute clearly and confidently.
- **Reflection on learning**: Teams discuss what worked, what didn't, and what they would improve next time.

Teamwork & Process

- **Regular and professional communication**: Teams check in frequently and adapt their plans as needed. Communication is about development, not getting an assignment done.
- **Issues are addressed early**: Problems are discussed and resolved before they escalate.
- **Clear task ownership**: Everyone knows what they are responsible for at each stage and this is visible in the project management board.

LinkedIn Project Report

- **Concrete examples**: Reports include specific stories of challenges faced and skills applied.
- **Professional framing**: The writing clearly signals readiness for graduate-level work.
- **Evidence of growth**: Students articulate how their capability changed during the project.

Final Note

Every major requirement in this specification maps directly to the marking rubric. Read this document alongside the rubric and use it as a checklist throughout the project.

This assignment is challenging by design. Treat obstacles as learning opportunities, work steadily, and focus on developing professional capability rather than simply completing tasks.

Marking Scheme

Ensure that you follow the processes and guidelines introduced in class to produce high quality work. This rubric provides you with the characteristics of exemplary to very limited work in relation to task criteria, based on the subject learning outcomes (SLO):

- SLO1 - Create websites using Content Management Systems, including creating content and custom templates or themes
- SLO2 - Select and apply DevOps practices for collaborative development and deployment
- SLO3 - Manage self-directed professional learning to develop skills and increase employability

The marks are split evenly between **product** (the website itself), and the professional development **process**.

Process-related criteria (documentation, version control, workflow, and project management) are assessed using observable evidence from the project repository, commit history, hosting environments, and project management tools. Markers look for evidence that development occurred progressively throughout the project rather than being assembled shortly before submission. Markers may consider the distribution and timing of commits when assessing version control and teamwork.

Project Group Component

Criteria	Exemplary (9, 10)	Good (7, 8)	Satisfactory (5, 6)	Limited (1-4)	Very Limited (0)
Publishing SLO2 10%	Site is available, up-to-date and working in production, staging and local environments. Local environment is demonstrated in project.html.	Site is available in production and local development, but not staging	Site is available in production and staging but not local development	Site is only available in local development or production	Site is not available anywhere
Content SLO1 5%	Site content is complete and accurate; text and media are all well-integrated.	Site content is mostly complete and accurate; most text and media are integrated appropriately with only minor gaps or inconsistencies.	Site content is present but uneven; some pages lack complete text or media; mostly well-integrated.	Site content is incomplete; not all media are present or not well-integrated.	Minimal or poor site content; not all media are present.
Goal-Driven SLO1 5%	Goals of the site are clear in the design and calls-to-action are well used.	Site design generally reflects its goals and audience; some calls-to-action are present but not consistently emphasised.	Some attempt is made to align design with site goals, but calls-to-action are weak, inconsistent, or unclear.	Design has some goal-orientation but mostly misses the opportunities to lead users; calls-to-action are not evident.	There is no sense of the site design being goal-driven.
Information Architecture SLO1 5%	Content has been thoughtfully organised to be suitable for site goals; navigation is intuitive.	Content organisation is generally logical; navigation is mostly intuitive with only minor issues.	Content organisation is somewhat inconsistent; navigation works but requires extra effort.	Not all content is well-organised; navigation is not always intuitive.	Content placement seems illogical; navigation is difficult.

Information Design SLO1 5%	Content is clearly formatted for readability and scanning; images support and enhance meaning.	Content is mostly well formatted and readable; images generally support the text.	Some formatting supports readability, though parts of the content remain dense or uneven. Images add limited value.	Formatting is inconsistent or does little to improve readability.	Little or no evidence of purposeful content formatting.
Interface Design SLO1 15%	Design is professional, visually consistent and well-suited for site goals and audience.	Design is mostly consistent and appropriate for the audience and goals, with only minor inconsistencies.	Design shows some coherence but includes noticeable inconsistencies or weak alignment with audience or goals.	Design is inconsistent or poorly suited to the audience and purpose.	Poor-quality design, not suitable for site goals and audience, or not using custom theme.
Theme SLO1 10%	Theme development is complete; structure and code follow standards; theme is reusable with no hard-coded specific content.	Theme is mostly complete and mostly follows standards with only minor issues.	Theme functions but shows multiple structural or standards issues, limiting maintainability or reusability.	Theme development is incomplete; problems with structure, standards, or reusability.	No custom theme developed.
Version Control SLO2 10%	Version control well-used with multiple commits by all team members, meaningful commit messages, no inappropriate files committed. Commit history shows iterative development.	Version control is used regularly during development. All members contribute and messages are generally clear. Commit history shows development over time with only minor inconsistencies.	Version control is used but inconsistently. Commit messages are sometimes vague and contributions from members are uneven.	Version control used minimally. Commits are infrequent, poorly described, or concentrated late in project. Commit history does not demonstrate meaningful development activity.	Version control not used at all
Project Management and Communication SLO2 15%	Project management tools are used consistently to organise and track work. Tasks are clearly defined, assigned, and updated throughout the project. The board shows evidence of planning, coordination, and progress over time.	Project management tools are used to organise work and track most tasks. Tasks are generally clear and progress is visible, though updates may be somewhat inconsistent.	Project management tools are used but inconsistently. Some tasks are defined or tracked, but planning and coordination are limited.	Project management tools are used minimally. Few tasks are defined or updated, and there is little evidence of coordinated planning.	No meaningful use of project management tools. Little or no evidence of planning or coordination. Or tools are not visible to markers.
Documentation SLO2 20%	Documentation is complete, accurate, well-written, and clearly supports development, deployment, and maintenance.	Documentation covers most required areas well and is generally clear, though some sections lack detail or clarity.	Documentation covers some expected areas but is incomplete, inconsistent, or difficult to follow.	Documentation is mostly incomplete or contains significant errors.	Documentation is mostly incomplete or incorrect

Project Individual Component - LinkedIn Project Report

LinkedIn Project Report SLO3	Report is well-written, goal-driven for employability, and effectively summarises the project, highlighting the skills used/learned and the role/s performed; correctly links to the rest of the team.	Report clearly describes the project and personal role with some examples of skills used or developed; links to team members are included.	Report summarises the project but provides limited detail about individual contributions or skill development.	Report is mostly complete but is not well-written, lacks promotion and/or doesn't highlight roles and other team members.	Report is mostly incomplete or incorrect.

Presentation - Group Component

Criteria	**Exemplary (9, 10)**	**Good (7, 8)**	**Satisfactory (5, 6)**	**Limited (1-4)**	**Very Limited (0)**
Content & Quality of Workflow SLO2 30%	Workflow integrates key professional practices including project management, version control, testing, and automation, with clear use of local, staging, and production. Strong understanding of deployment processes.	Workflow includes several professional practices such as project management, version control, testing, or automation, and uses local, staging, and production environments, though some aspects are limited or manual.	Workflow misses or poorly covers one or two of project management, version control, testing and automation; local, staging and production environments are used.	Workflow is basic or incomplete, with little integration of professional practices and missing or poorly implemented development environments.	Workflow is incomplete, missing most of the expectations; missing one or more of the expected environments.
Comparison of workflows SLO2 30%	Multiple workflows and tools are effectively compared, and decisions are clearly justified including evaluation, testing, overall rationale.	Multiple workflows and tools are compared; decisions are mostly justified including some coverage of evaluation, testing, overall rationale.	Some workflows and tools are compared; the decision is somewhat justified including some evaluation and overall rationale, minimal evidence of testing.	Some workflows and tools are compared; with minimal evaluation and justification.	Only one workflow considered; no or minimal comparison or justification.
Visuals and Demonstration SLO2 15%	Demonstrations are effective and well planned. Use of visuals enhances the presentation and meaningfully connects with the content being presented.	Demonstrations generally support the explanations; visuals are used but could be better integrated.	Demonstrations are present but not realistic or not well executed. Visuals don't always line up with the talk or the presentation lacks enhancing visuals.	Demonstrations or visuals are present but limited in clarity or effectiveness.	Demonstrations are missing or messy. Visuals are poor.
Structure SLO2 15%	Students present information in a logical sequence which audience can follow. Strong introduction and conclusion.	Presentation structure is mostly logical with a clear beginning and end.	Presentation sequence makes it harder for audience to follow than it should be. Introduction and/or conclusion are not strong.	Presentation structure is uneven or confusing; transitions or overall flow are weak.	Poor sequence of information means audience cannot understand presentation. Start and finish are very weak.
Group Work SLO2 *10%*	Presentation is shared well, each part logically flows, contributing to one coherent presentation.	Most members participate meaningfully; transitions between speakers are mostly smooth.	Presentation is not shared evenly, each person's part is separate, changeovers are a bit messy.	Participation is uneven; some members contribute considerably less or transitions are awkward.	Not all members contribute and/or sharing is problematic.

Presentation – Individual Component

Delivery Style SLO2	Speaker maintains good eye contact, seldom returning to notes. Clear voice and good pronunciation. Voice is dynamic and enhances meaning of presentation.	Speaker maintains regular eye contact, speaks clearly with minor reliance on notes. Delivery is mostly confident.	Some eye contact, reading for too much of presentation. Voice is too low or unclear. Tone is not interesting.	Frequent reading from notes, inconsistent clarity, low projection or monotone delivery.	Reading for majority of presentation with little to no eye contact. Unclear voice, incorrect pronunciation. Tone is monotonous or uninteresting.